



North South University

Department of Electrical & Computer Engineering

PROJECT TITLE:

**A Python-based Linux system monitoring and
diagnostics**

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SEC: 2

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GITHUB LINK: https://github.com/thexeroclub/OS_Project_CSE323

YOUTUBE VIDEO LINK: <https://youtu.be/JUkj0Jod2vc>

1. Project Overview:

SysPulse Ultra V9.1 is a Python-based **Linux** system monitoring and diagnostic application that provides real-time information about system performance and health.

It monitors CPU, memory, battery, storage, network, security, kernel, logs, packages, and containers through a structured terminal dashboard. The application also provides health scoring, automatic diagnosis, historical analysis, and report generation while using caching and concurrent processing for efficient performance.

2. Technical Implementation

- **Python:** Used to develop the monitoring, diagnostic, reporting, and dashboard system.
- **psutil:** Collects CPU, memory, battery, temperature, storage, network, process, and system information.
- **tabulate:** Creates structured tables for system and diagnostic data.
- **colorama:** Provides color-coded terminal output for better readability.
- **Linux Utilities:** Tools such as smartctl, nvme, systemctl, journalctl, ufw, apt, docker, and podman are used for additional system diagnostics.
- **ThreadPoolExecutor:** Runs multiple monitoring tasks concurrently to improve performance and responsiveness.
- **JSON/HTML:** Used with TXT files to generate reports of monitoring sessions.

3. Key Features

Real-Time CPU & Memory Monitoring:

Monitors CPU usage, individual cores, RAM, swap, and CPU frequency with visual health bars.

Battery Monitoring:

Displays battery percentage, charging status, remaining time, and battery health.

Thermal Monitoring:

Tracks temperature sensors and fan speeds to help identify overheating.

Storage & Disk I/O:

Shows storage capacity, free space, usage percentage, and real-time disk read/write speeds.

Network Monitoring:

Tracks download/upload speeds, packets, errors, IP addresses, and network latency.

GPU Monitoring:

Displays NVIDIA GPU temperature, usage, VRAM, and power information when supported.

Resource Hunter:

Identifies the top CPU- and RAM-consuming processes and detects zombie processes.

Linux Health Monitoring:

Checks failed system services and general Linux system stability.

Disk Health Diagnostics:

Uses SMART/NVMe tools to check disk health, temperature, wear, power-on hours, and errors.

Boot Health Analysis:

Analyzes boot time and identifies slow-starting system services.

Security Monitoring:

Checks firewall status, listening ports, SSH activity, authentication failures, and root processes.

Kernel & Log Analysis:

Monitors kernel status, errors, warnings, OOM events, and important system log entries.

System Health Score:

Combines multiple system conditions into an overall health score out of 100.

Automatic Diagnosis:

Identifies detected problems and provides recommendations for resolving them.

Package & Update Monitoring:

Checks available updates, broken packages, and whether a system reboot is required.

Container Monitoring:

Monitors Docker/Podman containers, including status, CPU, memory, and network usage.

Historical Performance:

Tracks CPU, RAM, latency, and health-score trends over the monitoring session.

Real-Time Alerts:

Generates warnings and critical alerts for high resource usage, temperatures, storage, and latency.

Webhook Notifications:

Can send system alerts to an external webhook without interrupting the dashboard.

Multi-Format Reports:

Automatically generates TXT and JSON reports, with optional HTML output, containing session statistics, alerts, health scores, and diagnostic results.

4. Deployment Strategy

SysPulse Ultra V9.1 is designed as a lightweight Linux command-line monitoring utility.

Lightweight Terminal Deployment:

Runs directly in the terminal with structured tables, progress bars, and color-coded information.

Configurable Execution:

Users can configure refresh rate, history size, diagnostics, alerts, and report formats through command-line options.

Continuous or Single Monitoring:

Supports both continuous real-time monitoring and single-snapshot diagnostics.

Graceful Compatibility:

Unavailable hardware, sensors, or Linux utilities are handled safely without stopping the application.

Efficient Diagnostics:

Expensive diagnostic operations are cached and refreshed at set intervals to reduce system resource usage.

5. Challenges & Difficulties Faced (STAR FORMAT)

During the development of SysPulse Ultra V9.1, several challenges were encountered and addressed using the STAR approach.

Challenge 1: System Performance

Situation: Running many monitoring and diagnostic operations could slow the dashboard.

Task: Keep the application responsive while collecting multiple system metrics.

Action: Used ThreadPoolExecutor for concurrent tasks and caching for expensive diagnostics.

Result: The dashboard remained responsive while performing extensive system monitoring.

Challenge 2: Hardware and Linux Compatibility

Situation: Some systems may not provide certain sensors, utilities, or hardware interfaces.

Task: Prevent the application from crashing when optional features are unavailable.

Action: Added availability checks, exception handling, and fallback values for unsupported components.

Result: The application continued running normally even when some diagnostic features were unavailable.

Challenge 3: Accurate Real-Time Measurements

Situation: Network and disk statistics are provided as cumulative counters rather than direct speed values.

Task: Calculate accurate real-time transfer rates.

Action: Compared previous and current counter values using the elapsed time between samples.

Result: Reliable network and disk read/write rates were displayed in real time.

Challenge 4: Understanding Overall System Health

Situation: The application collected a large amount of raw diagnostic information.

Task: Make the information easier for users to understand.

Action: Implemented health scoring, alerts, and automatic diagnosis based on defined thresholds.

Result: The system could provide an overall health score and actionable recommendations instead of only displaying raw data.

Challenge 5: Preserving Monitoring Data

Situation: Real-time information could be lost after the monitoring session ended.

Task: Save important session information for later analysis.

Action: Implemented session history and TXT, JSON, and optional HTML report generation.

Result: Monitoring results could be stored and reviewed after the application was closed.

6. Conclusion

SysPulse Ultra V9.1 expands traditional system monitoring into a comprehensive Linux health and diagnostic platform. The project combines real-time resource monitoring with hardware inspection, storage diagnostics, boot analysis, security visibility, kernel monitoring, journal analysis, package health, container monitoring, automated health scoring, diagnosis, alerting, historical tracking, and multi-format reporting.

The application's architecture emphasizes efficiency through concurrent task execution, controlled diagnostic intervals, caching, and graceful handling of unavailable system components. Its terminal-based dashboard provides a structured and visually accessible interface while maintaining the lightweight nature expected from a command-line system utility.

Overall, SysPulse Ultra V9.1 demonstrates how Python and Linux system interfaces can be integrated to create a practical operating-system monitoring and diagnostic solution. The project goes beyond simple CPU and RAM monitoring by providing users with a broader understanding of system health, potential failures, performance trends, security conditions, and actionable diagnostic recommendations.